28 When the number of protons and the number of neutrons in a nuclide are both “magic numbers”, it is more stable than expected. Such nuclides are termed “doubly magic”.

The first few “magic numbers” are 2, 8, 20, 28, 50, 82, and 126.

How many of the following five nuclides are “doubly magic”?

$^{28}_{8}O$	$^{40}_{20}Ca$	$^{56}_{26}Fe$	$^{50}_{28}Ni$	$^{126}_{50}Sn$

A 1 **B** 2 **C** 3 **D** 4

Ans: (B)

$^{28}_{8}O$: 8 protons, 20 neutrons (doubly magic)

$^{40}_{20}Ca$: 20 protons, 20 neutrons (doubly magic)

$^{56}_{26}Fe$: 26 protons, 30 neutrons (not)

$^{50}_{28}Ni$: 28 protons, 22 neutrons (not)

$^{126}_{50}Sn$: 50 protons, 76 neutrons (not)

[Turn over

29 Radon 222, ^{222}Rn decays to Lead 210, ^{210}Pb via a series of three alpha and two beta decays